Shell Script control statements (bash)

1. If…else
2. for
3. while
4. until
5. case

if statement

Syntax:

if [expression/control command] ;

then

statements

fi

- The Shell *expression* is evaluated in the above syntax. If the resulting value is *true*, given *statement(s)* are executed. If the *expression* is *false*, then no statement will be executed.

if statement

```
num=3

if [ $num -gt 5 ]
then
   echo "Greater"
else
   echo "Smaller"
fi



==================

if [ -f "file.txt" ]
then
   echo "File exists"
else
   echo "File not found"
fi
```

```
num=5

if [ $num -gt 5 ]
then
   echo "Greater than 5"
elif [ $num -eq 5 ]
then
   echo "Equal to 5"
else
   echo "Less than 5"
fi
```

While loop

```
while [ condition ]
do
   commands
done
```

```
i=1
while [ $i -le 5 ]
do
   echo $i
   i=$((i+1))
done
```

break

Example for break:

```
i=0;
while [ $i -lt 5 ]
do
        echo "Number: $i"
        i=`expr $i + 1`
if [ $i -eq 2 ];
then
        break
fi
done
echo "All Done!“
```

output:

```
Number: 0
Number: 1
All Done!
```

Continue

Example for continue:

```
i=0;
while [ $i -lt 5 ]
do
        if [ $i -eq 2 ];
        then
                i=`expr $i + 1`
                continue
        fi
 echo "Number: $i"
 i=`expr $i + 1`
 done
        echo "All Done!"
```

Output:
Number: 1
Number: 3
Number: 4
Number: 5
All Done!

For loop

```
for variable in list
do
   commands
Done
==================
for var1 in 1 2 3
do
      echo $var1
done
```

```
a="1 2 3 4 5"
for NUM in $a
do
      for NUM1 in $a
      do
            echo -n "*"
      done
      echo
done
```

For loop

```
for i in {1..5}
do
   echo $i
done
```

Until loop

```
until command
do
   Statement(s) to be executed until
command is true
done
```

```
a=0
until [ ! $a -lt 10 ]
do
echo $a
a=`expr $a + 1`
done
```

Case statement

```
ch=0
while [ $ch -le 5 ]
do
echo "1 : Addition"
echo "2 : subtraction"
echo "3 : multiplication"
echo "4 : division"
echo "5 : modulus"
echo "6 : exit"
echo "enter your choice:"
read ch
        if [ $ch -le 5 ] ; then
                echo -n "enter 1st no:"
                read no1
                echo -n "enter 2nd no:"
                read no2
        else
                echo "invalid choice"
        fi
clear
case "$ch" in
        1)
echo " addition "
echo " ans = " ` expr $no1 + $no2 ` ;;
2)
                echo " subtration "
                echo " ans = " ` expr $no1 - $no2
` ;;
        3)
                echo " multiplication "
                echo " ans = " ` expr $no1 \*
$no2 ` ;;
        4)
                echo " division "
                echo " ans = " ` expr $no1 /
$no2 ` ;;
        5)
                echo " modulus "
                echo " ans = " ` expr $no1 %
$no2 ` ;;
        6)
                exit
        esac
Done
```

- To execute expression
- `expr $var1 + $ var2`

Vi editor

- I for insert text – to go to insert mode
- Esc for going back to command mode

- : end
- w - to save
- q – quit

- :wq - Save and Quit

Command Line argument in Shell Script

Variable	Meaning
$0	Script name
$1	First argument
$2	Second argument
$3	Third argument
$#	Number of arguments
$*	All arguments (single string)
$@	All arguments (separately)
$?	Exit status of last command

```
echo "Script Name: $0"
echo "First Argument: $1"
echo "Second Argument: $2“

Jay$ ./test.sh Hello World
Script Name: ./test.sh
First Argument: Hello
Second Argument: World

==============================

sum=$(($1 + $2))
echo "Sum is: $sum"
```